IoT based patient health monitoring system ECG, Heart Rate, Temperature sensor

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Abstract:
The Internet of Things (IoT) has enabled efficient and real-time health monitoring for patients using smart and connected devices. This paper explores an IoT-based patient health monitoring system that measures and transmits health parameters such as ,heart rate, electrocardiogram (ECG), electrocardiogram (ECG) and body temperature [1], [2]. The sensor signals are received and processed by an ESP8266 NodeMCU microcontroller, which sends data to the ThingSpeak cloud platform [3]. A simple machine learning (ML) model is applied using MATLAB Analytics and Visualization in ThingSpeak [4] to predict abnormal readings. [4] The proposed system is portable ,low-cost, and suitable for remote patient monitoring applications.

Keywords--
IoT, Health Monitoring, ECG Sensor, Heart Rate, Temperature Sensor, ThingSpeak, Machine Learning.

Introduction:

Real-time health monitoring has become increasingly important in modern healthcare. Existing monitoring techniques requires manual measurements and frequent hospital visits, which can delay the treatment [1]. Using IoT technology , the system automatically collects and analyzes patients’ key body paramters such as ECG, temperature and heart rate.These readings are wirelessly transmitted to a cloud platform, ThingSpeak, enabling doctors or caregivers to view them remotely [3]. Machine learning algorithms used here, can analyze the stored data and predict possible abnormalities [4], enhancing patient safety.

Components:

The proposed system includes the following modules [1]:

1. ESP8266 NodeMCU**:** Microcontroller with in-built Wi-Fi capability.
2. ECG Sensor: Measures heart’s electrical activity. [2]
3. Heart Rate Sensor**:**Monitors pulse rate in BPM.
4. Temperature sensor**:** Monitors body temperature and humidity.

5. ThingSpeak IoT Platform [3]: Used as the cloud dashboard to collect, store, and visualize real-time data from each sensors. Each sensor's readings are uploaded to separate ThingSpeak fields and can be monitored via web or mobile.

Existing system:

Traditional health monitoring systems mainly use wired connections. They often lack cloud storage and connectivity and do not include predictive analysis [1].Most existing systems only record data from the sensors but do not provide any early alerts or ML-based prediction analysis. [4]

Developed system:

The implemented system aims to provide continuous health monitoring of patients by including multiple bio sensors with an IoT platform and machine learning for prediction analysis [2], [4]. [4] The system consists of three sensors as its input layer : the ECG sensor (AD8232) for cardiac signal measurement, the heart rate sensor for pulse detection, and the DHT11 temperature sensor for monitoring body temperature. [3] .At the core of the system is the ESP8266 NodeMCU microcontroller, which functions as the central processing unit and communication unit. All sensor readings are collected and processed by this ESP8266. The collected data are transmitted to the ThingSpeak cloud through the in-built Wi-Fi, where real-time visualization of the patient's physiological parameters is performed [3].The ThingSpeak's MATLAB Analytics module is used to execute machine learning algorithms such as K-means clustering and linear regression to predict abnormal trends [4].When abnormal values are detected, a React–ThingHTTP mechanism triggers an automated alert via a webhook or notification system. This allows caregivers to receive immediate updates regarding the patient's condition. The proposed design emphasizes low-cost implementation, portability, making it suitable for both hospital environments [1].

System Overview:

This IoT-based patient health monitoring system is designed to efficiently collect, process, and analyze physiological data in real time. [2]

The hardware and software components are as follows:

1. ECG Sensor (AD8232**):** Measures the electrical activity of the heart and provides output signals for heart rate monitoring.
2. Heart Rate Sensor**:** Detects pulse signals and calculates beats per minute (BPM).
3. DHT11 Temperature Sensor**:**
This sensor, measures the patient's body temperature and sends data in digital form.
4. ESP8266 NodeMCU**:** Acts as the microcontroller and Wi-Fi module for transmitting sensor data to the cloud.
5. ThingSpeak IoT Cloud**:** Receives, stores, and visualizes the received data. It also integrates MATLAB Visualization for implementing simple machine learning models.

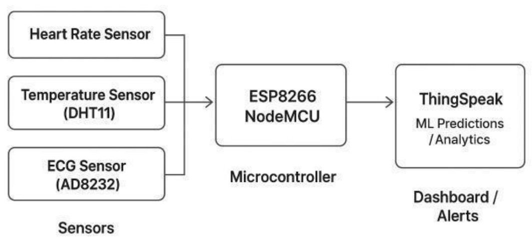

Fig.1 Block diagram of the project

(Fig. 1) shows the connections between sensors, the ESP8266, and ThingSpeak cloud platform [3]. The microcontroller receives sensor data, uploads it to ThingSpeak, and triggers notifications via the React and ThingHTTP modules [3] when abnormal conditions are detected.

Working Principle:

The proposed IoT-based health monitoring system operates by continuously sensing and transmitting health parameters of the patient. [2] The DHT11 sensor measures body temperature, while the AD8232 ECG sensor records the heart's electrical activity and the pulse sensor records the pulse rate. [1] These analog readings are processed by the NodeMCU (ESP8266) microcontroller, which converts them into digital form and uploads the data to the ThingSpeak cloud platform [3] through Wi-Fi. [3] The uploaded data is visualized in real time on the ThingSpeak dashboard. Machine learning algorithm such as K-Means clustering and linear regression [4] are applied to predict abnormal variations in temperature or heart rate. When the predicted value crosses a predefined threshold, a notification or alert is automatically triggered using ThingHTTP and React services in ThingSpeak [3].

This ensures that any critical change in the patient's condition is detected early, allowing timely medical intervention.

Results:

The ECG sensor (AD8232) provides analog waveform signals representing the electrical activity of the heart, while the heart rate sensor measures beats per minute (BPM), and the DHT11 sensor records temperature. These values were displayed as live graphs on ThingSpeak, enabling continuous observation [3] of patient health.To enhance system intelligence, a linear regression model is applied to temperature data to predict the next possible reading. [4] This model uses the relationship between previous temperature samples and time to predict future values. When the predicted reading exceeded the normal physiological range, the system could be configured to issue an alert.

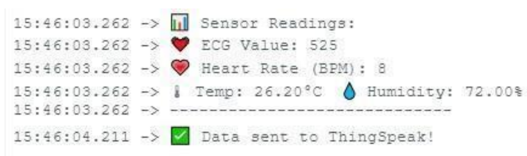

Fig.2 shows the output from the sensor configurations.

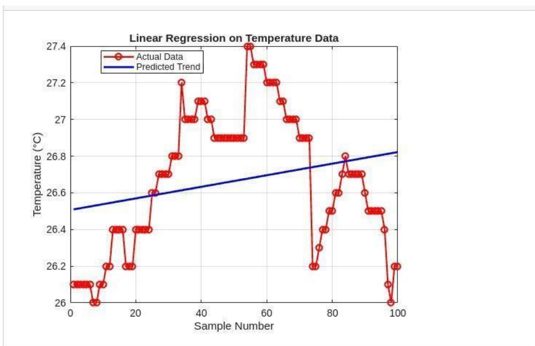

Fig.3 shows the Linear regression model of the temperature sensor.

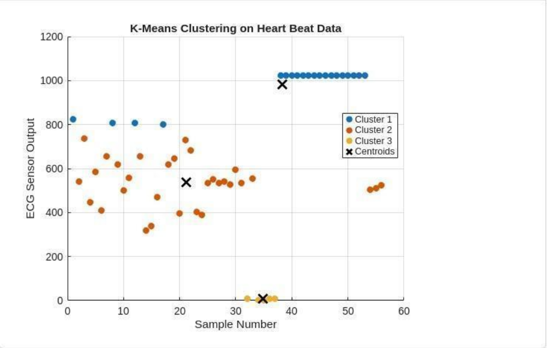

Fig.4 shows the K-means clustering of the heart rate sensor.

Conclusion:

The IoT-enabled health monitoring system” is developed to record ECG, heart rate, and temperature using the ESP8266 NodeMCU. [1] Data collected from the sesnors are transferred to ThingSpeak for continuous observation and evaluation. [3] The system integrates ThingSpeak to automatically detect abnormal sensor readings and trigger alerts to caregivers. [3] The cloud platform enables data visualization, trend prediction, and real-time notification using basic machine learning alogorithms such as K-means clustering and linear regression. Although the actual physical output has not yet been tested with live sensors and patients, the simulation and theoretical results show that the system is capable of functioning as intended. The sensors collect data and ThingSpeak stores and visualizes it , and the in-built machine learning algorithms (K-means clustering and linear regression) predicts abnormal readings.

Future Scope:

In the future, this system can be enhanced by integrating additional sensors such as blood pressure, oxygen saturation (SpO_2), and respiration rate for more comprehensive patient monitoring. [1] A mobile application or web dashboard can be developed for real-time access and control by healthcare professionals. Furthermore, advanced machine learning and AI models can be employed to predict medical conditions more accurately and provide preventive alerts to doctors and family members. [4]

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